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Effects of Physical Exercise on the Expression of MicroRNAs: A Systematic Review

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Abstract

Silva, FCd, Iop, RdR, Andrade, A, Costa, VP, Gutierres Filho, PJB, and Silva, Rd. Effects of physical exercise on the expression of microRNAs: A systematic review XX(X): 000–000, 2019—Studies have detected changes in the expression of miRNAs after physical exercise, which brings new insight into the molecular control of adaptation to exercise. Therefore, the objective of the current systematic review of experimental and quasiexperimental studies published in the past 10 years was to assess evidence related to acute effects, chronic effects, and both acute and chronic effects of physical exercise on miRNA expression in humans, as well as its functions, evaluated in serum, plasma, whole blood, saliva, or muscle biopsy. For this purpose, the following electronic databases were selected: MEDLINE by Pubmed, SCOPUS, Web of Science, and also a manual search in references of the selected articles to April 2017. Experimental and quasiexperimental studies were included. Results indicate that, of the 345 studies retrieved, 40 studies met the inclusion criteria and two articles were included as a result of the manual search. The 42 studies were analyzed, and it can be observed acute and chronic effects of physical exercises (aerobic and resistance) on the expression of several miRNAs in healthy subjects, athletes, young, elderly and in patients with congestive heart failure, chronic kidney disease, diabetes mellitus type 2 associated with morbid obesity, prediabetic, and patients with intermittent claudication. It is safe to assume that miRNA changes, both in muscle tissues and bodily fluids, are presumably associated with the benefits induced by acute and chronic physical exercise. Thus, a better understanding of changes in miRNAs as a response to physical exercise might contribute to the development of miRNAs as therapeutic targets for the improvement of exercise capacity in individuals with any given disease. However, additional studies are necessary to draw accurate conclusions.

Key Words: endurance, resistance, rehabilitation

Introduction

Physical exercise is a feasible alternative for the nonpharmacological treatment of several diseases. Nevertheless, despite its financial feasibility (29) and benefits, including reductions in the development and severity of cardiovascular, metabolic, rheumatologic, pulmonary, oncological, and neurological diseases (2,3,20,23,43,44,64,97), physical exercise is still poorly used in health care treatment. Furthermore, through inducing functional changes in several physiological systems, physical exercise is considered an important promoter of improved quality of life (13,69).

Many of these changes have been attributed to exercise-induced epigenetic alterations that alter the expression profile of several genes (8). Generally speaking, epigenetic changes on the DNA or chromatin structure are able to influence the transcription of genes, independently of their primary sequences. The most common exercise-induced epigenetic changes involve histone methylation and acetylation, DNA methylation, and the expression of various microRNAs (miRNAs) (69).

miRNAs are small noncoding RNAs involved in epigenetic regulation. Specifically, these molecules modulate the repression of post-transcriptional gene expression by attenuating or silencing protein translation (34,45). According to Drummond (34), a single miRNA can have several target genes, and several miRNAs might affect a single transcript, thus rendering a relatively simple process rather complicated. As reported by Xu et al. (100), to date, more than 2,000 miRNAs have been identified in the human genome. Of note, approximately 60% of all proteins can be modulated by miRNAs.

Studies have detected changes in the expression of miRNAs after physical exercise, which brings new insight into the molecular control of adaptation to exercise. Aoi et al. (4) and Baggish et al. (6) have shown that physical exercise is a potent gene expression activator, with considerably variable expression patterns that depend on the type of exercise. Furthermore, exercise, especially resistance exercise, results in the activation of several signaling cascades, which, in turn, promote an increase in protein synthesis (32,57,95).

Some of these changes in gene expression might be attributed to changes in miRNA levels. Specifically, some studies have shown that exercise induces changes in miRNA levels within different cell types (6,28,53,86), and fluctuations in miRNA levels can increase the adaptations of training by regulating specific genes involved in these adaptations (49).

Evidence suggests that miRNAs could be used as biomarkers in the diagnosis, treatment, and prognosis of many diseases

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(12,19,27,42,47,73,103). In addition to the changes resulting from pathological conditions, miRNAs might also undergo changes due to physical exercise, thus potentially contributing to the beneficial effects observed with physical exercise (55,85). Although most miRNAs are located within cells, they have also been consistently identified in bodily fluids, in which case they are referred to as circulating miRNAs (ci-miRNAs) (35).

To date, the review articles available, involving humans and animals, summarize recent developments in the field of ci-miRNAs as a response to exercise, including future perspectives and their application as a new class of biomarkers of physical performance and adaptive capacity to training, and its relevance to chronic diseases and prevention of degenerative diseases. For example, revisions published in 2015 included 9, 14, 9, and 8 primary articles (1,36,40,56,100); em 2016 included 26 and 16 primary articles (56,74); and in 2017 included 30 articles on exercícios/treinamento and ci-miRNAs (84). A previous review article (52) presents a complete overview of the main available knowledges about the preanalytical management of the sample for ci-miRNA evaluation along with the importance of miRNA as regulators of the response to physical activity and their possible future use in antidoping settings. Therefore, the objective of the current systematic review was to assess the evidences relating to effects acute, chronic, and acute and chronic of physical training on miRNA expression in human evaluated in serum, plasma, whole blood, saliva, or muscle biopsy and their functions, of experimental and quasiexperimental studies published in the past 10 years.

Methods

This systematic review was registered under the CRD42016032770 code, in the International Prospective Register of Systematic Reviews—PROSPERO, and it followed the proposed Preferred Reporting Items for Systematic Review and Meta-analyses: The PRISMA Statement (61). It was approved by Santa Catarina State University.

Eligibility Criteria

Studies were eligible for further analysis if the following inclusion criteria were met: (a) experimental and quasiexperimental studies; (b) focusing miRNAs and physical exercise; (c) including different human groups (male and female; individuals healthy and with pathologies; athletes and nonathletes; and young and old individuals); (d) compared with another intervention or a nonexposed control group studies; (e) indexed in previously selected databases; (f) with abstracts were available; (g) with permissions from access to full articles; (h) which could be accessed online in the past 10 years; and (i) there were no language restrictions. Exclusion criteria included the following: (a) review articles; (b) congress abstracts; (c) studies using animal models; and (d) did not address the effect of physical exercise.

Search Strategy

The selected electronic databases were *MEDLINE (Medical Literature Analysis and Retrieval System on-line)* via *PubMed*, *SCOPUS (Elsevier), Web of Science*, and also a manual search in references of the selected articles. In the search strategy, keywords were selected according to the Medical Subject Headings (MeSH) database (Table 1). All search strategies were performed during April 2017.

The titles and abstracts of all articles identified with the search strategy were evaluated independently by 2 authors of this study. Similarly, each of the above authors evaluated the full articles and made their selections according to the previously specified eligibility criteria. There was no disagreement between the reviewers.

Data Extraction

The extracted data included the following: identification of the article (first author and year of publication), study design, studied population (sample size, participants' characteristics including sex and mean age), sample type, analyzed miRNAs, type of physical exercise, correlation between miRNA and physical exercise, and function. It was considered an acute effect, those studies that analyzed the change in miRNA expression after a single exercise session. Already the chronic effect, those studies that performed an exercise program for a specific week period.

Quality Assessment

Risk of bias across studies was evaluated through careful evaluation of informations withdrawn mainly from sections: (a) methods; (b) results, and (c) conclusions, when applicable, of the respective studies. Risks of bias were critically compared with other similar studies and with the specialized literature and are exposed throughout the discussion section of this article.

Results

The search yielded 345 articles, 82 of which were excluded as duplicates, and 209 were excluded for having titles and abstracts that did not comply with the eligibility criteria. No articles were selected from the gray literature. After a detailed evaluation, 40 studies were considered to be potentially relevant and were included in the review. Two articles were included as a result of the manual search, totaling 42 studies included in the review corresponding to 12%. Figure 1 illustrates the flowchart of this process.

The main characteristics of the included studies are described in Tables 2, 3, 4, and 5. The majority were performed in the United States (4,6,33,54,60,75,77–79,104), Australia (11,21,30,37,71,82,90,102), and Germany (5,22,80,89,91). Sample sizes varied between 7 (11,33,104) and 59 (91), and the studied populations were healthy people, athletes, young, and old individuals. Only 7 studies involved individuals with pathologies, including chronic heart failure (CHF) (80,101), chronic kidney disease (CKD) (93), type 2 diabetes mellitus (DM2) associated with morbid obesity (81), prediabetic (72), and intermittent claudication (26,68). Most of the studies were on male subjects only. The mean age of participants varied between 19.1 (6) and 80.1 years (63). The most common sample type was muscle biopsy (vastus lateralis) (11,28,33,37,46,60,63,65,70,81,82,102) and of plasma (6,22,24,25,41,66,71,72,90,91,93,104).

A variety of miRNAs were assessed related to inflammatory processes; vascular health; immune system; disease associated biomarkers; markers anaerobic capacity; anti-inflammatory and protective reaction; myoblast proliferation regulate hematopoietic; apoptosis; immune function; transcription regulation and membrane traffic of proteins; muscle lesion; skeletal muscle adaptation; angiogenic mediator; prognostic markers of cardiovascular outcomes; declines in skeletal muscle mass; increased fat mass and anabolic resistance; gain of muscle mass; cardiovascular diseases, inflammation and angiogenesis; atherogenesis and endothelial dysfunction; and control and repair of immune

Table 1
Descriptors used in search strategy.*

Headings	Keywords
Physical exercise	"Physical Therapy Modalities"[Mesh], "Physical Therapy Modalities," "Modalities, Physical Therapy," "Modality, Physical Therapy," "Physical Therapy Modality," "Physiotherapy (Techniques)," "Physiotherapies (Techniques)," "Physical Therapy Techniques," "Physical Therapy Technique," "Techniques, Physical Therapy," "Exercise Movement Techniques"[Mesh], "Exercise Movement Techniques," "Movement Techniques, Exercise," "Exercise Movement Technics," "Exercise Therapy"[Mesh], "Exercise Therapy," "Therapy, Exercise," "Exercise Therapies," "Therapies, Exercise," "Exercise"[Mesh], "Exercise," "Exercises," "Exercise, Physical," "Exercises, Physical," "Physical Exercise," "Physical Exercises," "Exercise, Isometric," "Exercises, Isometric," "Isometric Exercises," "Isometric Exercise," "Exercise, Aerobic," "Aerobic Exercises," "Exercises, Aerobic," "Aerobic Exercise," "Resistance Training"[Mesh], "Resistance Training," "Training, Resistance," "Strength Training," "Training, Strength," "Weight-Lifting Strengthening Program," "Strengthening Program, Weight-Lifting," "Strengthening Programs, Weight-Lifting," "Weight Lifting Strengthening Program," "Weight-Lifting Strengthening Programs," "Weight-Lifting Exercise Program," "Exercise Program, Weight-Lifting," "Exercise Programs, Weight-Lifting," "Weight Lifting Exercise Program," "Weight-Lifting Exercise Programs," "Weight-Bearing Strengthening Program," "Strengthening Program, Weight-Bearing," "Strengthening Programs, Weight-Bearing," "Weight Bearing Strengthening Program," "Weight-Bearing Strengthening Programs," "Weight-Bearing Exercise Program," "Exercise Program, Weight-Bearing," "Exercise Programs, Weight-Bearing," "Weight Bearing Exercise Program," "Weight-Bearing Exercise Programs"
MicroRNAs	"MicroRNAs"[Mesh], "MicroRNAs," "MicroRNA," "miRNAs," "Micro RNA," "RNA, Micro," "miRNA", "PrimaryMicroRNA," "MicroRNA, Primary," "PrimarymiRNA," "miRNA, Primary," "pri-miRNA," "primiRNA", "RNA, Small Temporal," "Temporal RNA, Small," "stRNA," "Small Temporal RNA," "pre-miRNA," "premiRNA

*Source: author's own production.

function, cell cyte process, development, and growth. The types of physical exercise were endurance (bicycle ergometer, running, and rowing) and resistance training (leg extension and leg press). In addition, the effects of physical exercise on the analyzed miRNAs were acute effect (5,7,11,21,24–26,33,37,41,50,60,76–79,85,89,91,101,102), chronic effect (6,28,30,31,46,63,65,68,71,72,80,81,90,104), and acute and chronic effects (4,22,66,70,82,93). As for the correlation between physical exercise and miRNAs, 104 exhibited upregulation, whereas 100 exhibited downregulation after acute exercise. Furthermore, chronic physical exercise promoted upregulation of 40 and downregulation of 32 miRNAs. The effect acute and chronic physical exercise promoted upregulation of 24 miRNAs and downregulation of 25 miRNAs.

Discussion

The main objective of this study was to assess the evidences relating to effects acute, chronic, and acute and chronic of physical training on miRNA expression in human evaluated in serum, plasma, whole blood, saliva, or muscle biopsy and their functions, of experimental and quasiexperimental studies published in the past 10 years. A total of 42 studies were analyzed and revealed both acute and chronic effects of physical exercise (endurance and resistance) on the expression of several miRNAs in athletes, healthy young and older individuals, and patients with CHF, CKD, DM2 associated with morbid obesity, prediabetic, and intermittent claudication; most of the studied population were males.

Physical exercise is recognized as one of the main components in the promotion of health and the prevention and even treatment of several diseases. A single exercising session was found to be potentially beneficial on the expression of miR-1, -133a, -133b, -206, -499, -181a-5p, -16, -126, -532-5p, 15a, -96, -30e, -181a, -197, -130a, 151-5p, -221, -23b, -29b, -660, -140-5p, -338-, -let-7e, -199a-3p, -223, -652, -363, -181b, -1225-5p, -939, -125a-5p, -145, and -940.

The above results indicate that these miRNAs might partially reflect exercise-induced cardiac or skeletal muscle responses, or a temporary regulation mediated by miRNAs during exercise. Similarly, the reduction in miR-16 levels suggests that nonmuscle tissue is also necessary to cope with this type of stress (76).

A lower ci-miR-1 level after high-intensity interval exercise compared with vigorous-intensity continuous exercise (VICE) was identified by study of Cui et al. (25) suggesting the sprint intervals may induce more or different muscle damage than the sustained endurance exercise. In study by Guescini et al. (41), no significant changes were found on the expression of miR-1, -133a, -499, and -206 in extracellular vesicles (EVs) isolated from plasma of healthy men after a 40-minute running on a treadmill at 80% $\dot{V}O_2$ max. These findings suggest that circulating miRNA levels are modulated in response to exercise, and miRNAs may play important roles in exercise-induced adaptations such as gain of muscle mass. Guescini et al. (41) found increased ci-miR-181a-5p; thus, the authors propose EVs as a novel means of muscle communication potentially involved in muscle remodeling and homeostasis.

According to Gomes et al. (40), there is a consensus that miRNAs can play a role in muscle development and muscle response to stress, particularly myomiRs. miR-206, for example, stimulates myogenesis by negatively regulating inhibitors of MyoD in cultured C2C12 myoblasts (48). During muscle development in vitro, miR-1 promotes myogenesis by targeting an inhibitor of muscle gene expression, histone deacetylase 4 (40), in addition, presumably miR-1 regulating IGF-1 signaling (102). Mueller et al. (63) reported a downregulation of miR-1 associated with increased IGF-1 expression in both modalities (resistance exercise and eccentric ergometer sessions). The miR-499 is encoded in the myosin gene *Myh7* (94). *Myh7* encodes the β-MHC protein that is expressed highly in slow-twitch skeletal muscle. The miR-499 has been shown experimentally to regulate the expression of transcriptional repressors (*Sox6, Purβ*, and *Sp3*) of β-MHC expression. Multiple lines of evidence indicate that Sox6 is a major repressor of β-MHC expression in skeletal muscle. In addition to regulating fiber type, miR-499 have been

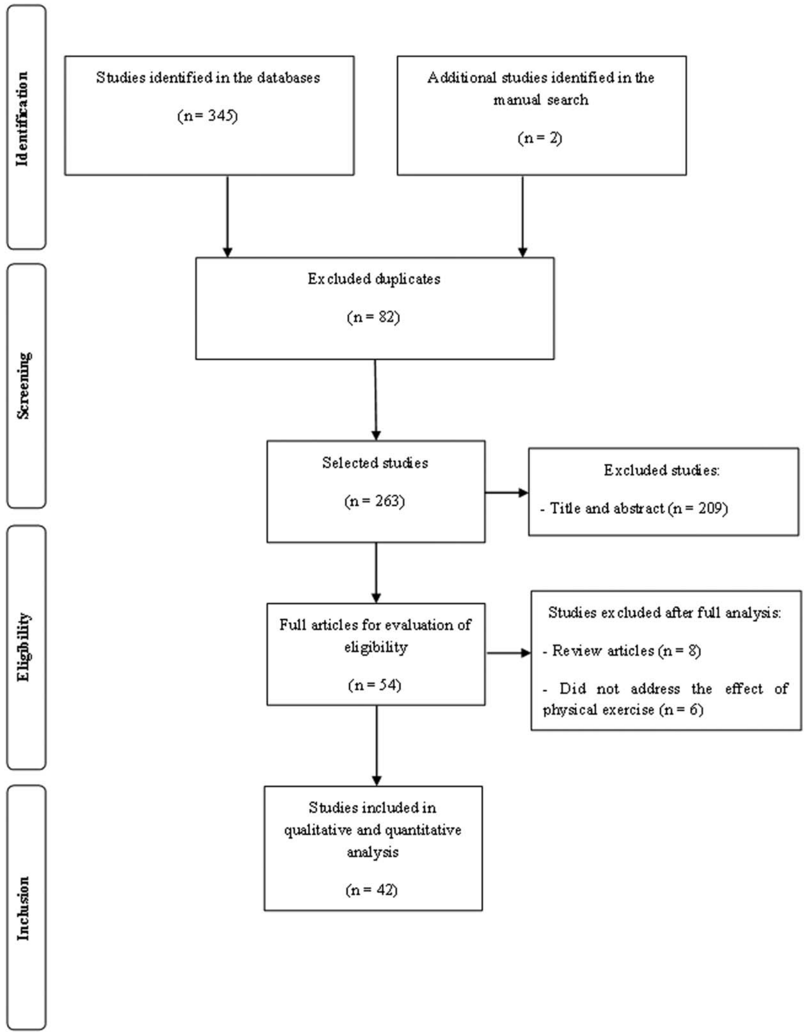

Figure 1. Study identification and selection process.

shown to regulate myostatin (*Mstn*) expression, suggesting that the myomiR network may coordinate changes in fiber type and muscle mass (58).

In the study of Tonevitsky et al. (89), there was an increase in miR-181a-5p levels after exercise and decreased in the relaxation period. The mRNAs ROPN1L (adjusted p-value of 0.00024) and SLC37A3 (adjusted p-value of 0.0019) were previously validated to be targets for miR-181a-5p and demonstrated pronounced anticorrelation with the miRNA expression profile. The *ROPN1L* gene encodes a member of the ropporin family. The encoded protein is involved in the targeting toward specific physiological substrates of protein kinase A, regulating glycogen, sugar, and lipid metabolism (14). The SLC37A3 protein belongs to transmembrane sugar transporters and is responsible for sugar metabolism (9). The miR-181a-5p is known to regulate hematopoietic lineage differentiation (15) and to modulate T cell sensitivity and selection (51).

In the study of Da Silva et al. (26) was identified that the expression of ci-miR-126 increased in circulation 30 minutes after exercise in the placebo session. The authors concluded that miR-126 signaling is redox-sensitive in patients with intermittent claudication. Uhlemann et al. (91) observed significant changes in circulating miR-126 levels in healthy individuals after different types of physical exercise. The levels of miR-126, an indicator of endothelial lesions, were increased after a single maximal symptom-limited exercise test and after 4 hours of cycling. After a marathon, miR-126 levels were significantly increased. According to the authors, maximal symptom-limited exercise tests and resistance training cause endothelial damage, whereas running a marathon induces damage to skeletal muscle cells, and resistance training only affects muscles. Twenty minutes of exercise on bicycle ergometer (i.e., the ten 2-minute bouts) with 1-minute rest interval between each bout of exercise reduced the expression level of miR-126 in healthy males (76). Thus, circulating miRNAs might serve as indicators of muscle lesions. According to Wardle et al. (98), certain c-miRNAs differ between endurance and strength and thus have potential as useful biomarkers of exercise training or play a role in exercise mode–specific training adaptations.

The miR-532-5p exhibited similar responses in study by McLean et al. (60) and by Radom-Aizik et al. (79) miR-532-5p levels were found to be increased in muscle tissue from healthy subjects after one session of aerobic exercise at 70% HRmax (60) and were found to be increased in peripheral blood—monocytes from healthy males after one session of bicycle ergometer (20′ at 82% $\dot{V}O_2$max).

Table 2
Synthesis of the common findings for each type/modality of exercise—acute effect (upregulation and downregulation).*†

			Correlation with physical exercise	
Studied population	**Sample type**	**Function**	**Upregulation miRNA (References)**	**Downregulation miRNA (References)**
Endurance				
Bicycle ergometer				
Healthy males	PBMCs (RT-PCR)	Inflammatory processes	miR-15a (77); miR-140-5p (77); miR-181a (77); miR-181b (77); miR-197 (77); miR-338-3p (77); miR-363 (77); miR-939 (77); miR-125a-5p (77); and miR-1225-5p (77)	let-7e (77); miR-23b (77); miR-130a (77); miR-145 (77); miR-151-5p (77); miR-221 (77); and miR-652 (77)
Healthy males	Neutrophilis (RT-PCR)	Inflammatory processes	miR-181b (75); miR-223 (75); miR-1225-5p (75); miR-939 (75); miR-125a-5p (75); and miR-940 (75)	miR-16 (75); miR-126 (75); miR-96 (75); miR-145 (75); miR-130a (75); miR-221 (75); miR-660 (75); miR-363 (75); and miR-151-5p (75)
Healthy males	Peripheral blood—monocytes (RT-PCR)	Vascular health	miR-15a (75); miR-30e (75); miR-660 (75); miR-140-5p (75); miR-29b (75); miR-338-3p (75); and miR-532-5p (75)	miR-23b (75); miR-130a (75); miR-151-5p (75); miR-199a-3p (75); and miR-221 (75)
Healthy males	Peripheral blood (NK cells) (RT-PCR)	Health promotion—immune system	miR-29b (78); miR-30e (78); miR-338-3p (78); and miR-363 (78)	let-7e (78); miR-126 (78); miR-130a (78); miR-151-5p (78); miR-199a-3p (78); miR-221 (78); miR-223 (78); and miR-652 (78)
Elite endurance athletes and moderately active controls	Blood (microarray and RT-PCR)	Disease-associated biomarkers	miR-30e (5); miR-30e (5); and miR-197 (5)	miR-181a (5)
Running on a treadmill				
Healthy male students	Plasma (RT-PCR)	Markers anaerobic capacity	miR-133b (41) and miR-181a-5p (41)	miR-1 (41); miR-133a (41); miR-206 (41); miR-499 (41); and miR-16 (41)
Athletes	Venous blood	Anti-inflammatory and protective reaction, myoblast proliferation regulate hematopoietic		miR-181a-5p (89)
Sprint interval cycling				
Healthy young men	Plasma (RT-PCR)	Markers anaerobic capacity	miR-499 (24)	miR-1 (24); miR-133a (24); miR-133b (24); miR-206 (24); and miR-16 (24)
High-intensity exercise X vigorous-intensity continuous exercise				
Healthy young men	Plasma (RT-PCR)	Markers anaerobic capacity	miR-1 (25); miR-133a (25); miR-133b (25); and miR-206 (25)	
Running				
Healthy males	Peripheral blood (RT-PCR)	Apoptosis, immune function, transcription regulation, and membrane traffic of proteins	miR-15a (21) and miR-96 (21)	
Maximal symptom—limit exercise test (G1) or bicycling (G2) or running a marathon (G3) or bicycle ergometer (G4)				
Healthy adults	Plasma (RT-PCR)	Muscle lesion	miR-126 (91)	
Maximal exercise test				
Male patients IC	Blood	Angiogenic mediator	miR-126 (26)	
Aerobic exercise				
Healthy individuals	Muscle biopsy (vastus lateralis)	Skeletal muscle adaptation	miR-532-5p (60)	
Incremental cardiopulmonary exercise test on a bicycle ergometer				
Males CHF	Serum (RT-PCR)	Prognostic markers of cardiovascular outcomes	miR-940 (101)	
High-intensity interval training (HIT) or moderate-intensity continuous training (MICT)				
Healthy males	Muscle biopsy (vastus lateralis)	Skeletal muscle adaptation		miR-486 (37)

Table 2
Synthesis of the common findings for each type/modality of exercise—acute effect (upregulation and downregulation).*† *(Continued)*

			Correlation with physical exercise	
Studied population	**Sample type**	**Function**	**Upregulation miRNA (References)**	**Downregulation miRNA (References)**
Single-leg extension exercise				
Healthy young and healthy elderly males	Muscle biopsy (vastus lateralis)	Skeletal muscle mass		miR-100-5p (102)

*NK = natural killer; RT-PCR = reverse transcription polimerase chain reaction; G1 = group 1; G2 = group 2; G3 = group 3; G4 = group 4; PBMCs = Peripheral blood mononuclear cells; IC = intermittent claudication; CHF = congestive heart failure.
†Source: Author's own production.

Chilton et al. (21) found increases in miR-15a. The same result was found in the studies Radom-Aizik et al. (79) and Radom-Aizik et al. (77). Chilton et al. (21) also observed an increase in miR-96 while Radom-Aizik et al. (76) observed a decrease. Other miRNAs also presented different alterations in responses to different modalities, intensities of physical exercises, and sample type analyzed, such as miR-181a (5,76,77), miR-660 (76,79), miR-223 (76,78), and miR-363 (76–78).

The upregulation was found in the miR-30e, miR-29b (78,79), miR-140-5 (77,79), miR-338-3p (78,79), miR-181b, miR-1225-5p, miR-939, miR-125a-5p (76,77), and miR-940 (76,101). The downregulation was found in the expression of miR-130a, miR-151-5p (76–79), miR-221 (77–79), miR-23b (77,79), let-7e (77,78), miR-199a-3p (78,79), miR-652 (77,78), miR-145 (76,77), miR-486, and miR-100-5p (37,102).

In addition to samples from muscle tissue (28,46,63,65,81,104), the identification of circulating miRNAs specifically regulated by exercise could provide new insight into the physiological adaptation to physical exercise. Denham et al. (30) found reduced miR-210 and increased miR-21 levels in blood, causing a cascade effect on the expression of the mature microRNA involved in cardiovascular function. Four differentially expressed miRNAs in peripheral blood mononuclear cell of healthy untrained policemen recruits were upregulated miR-21 after 18 weeks of running (31). The miR-21 is linked to vascular inflammation and skeletal and cardiac muscle function (88,92).

In a study on athletes before and after rowing, Baggish et al. (6) observed an increase in plasma levels of miR-146a, -222, -210, and -221, which have previously been implicated in inflammation, angiogenesis, and muscle adaptation to hypoxia and ischemia, thus suggesting the potential value of circulating miRNAs as biomarkers for exercise and their role as possible physiological mediators of exercise-induced adaptation. In addition, a positive correlation between increased miR-146a levels and the aerobic performance parameter $\dot{V}O_2max$ was observed. Furthermore, the different doses of acute aerobic exercise induced a distinct and specific c-inflammamiR response, which may be associated with control of the exercise-induced inflammatory cascade (5). Furthermore, Nowak et al. (68) found no significant changes in the levels of analyzed miRNAs from isolation of peripheral blood total nucleated cells and sera of patients with intermittent claudication after 12 weeks of walking on a treadmill. However, this result was not discussed by the authors.

In another study, healthy males submitted to 12 weeks of endurance exercise (bicycle ergometer at 55–91% Pmax) exhibited reduced levels of miR-1, -133a in a quantification of muscle-specific miRNAs (myomiRs) from the vastus lateralis. MyomiRs returned to

Table 3
Synthesis of the common findings for each type/modality of exercise—chronic effect (upregulation and downregulation).*†

Studied population	**Sample type**	**Function**	**Upregulation miRNA (references)**	**Downregulation miRNA (references)**
Endurance				
Running				
Healthy untrained policemen	PBMCs (RT-PCR)	Control and repair of immune function, cell cyte process, development, and growth	miR-21 (31)	
Treadmill				
Healthy young males	Blood	Cardiovascular diseases, inflammation, and angiogenesis	miR-21(53)	miR-210 (30)
Bicycle ergometer				
Healthy males	Muscle biopsy (vastus lateralis)	Skeletal muscle adaptation		miR-133a (65)
Resistance exercise training or eccentric ergometer training				
Elderly men and elderly women	Muscle biopsy (vastus lateralis)	Gain of muscle mass		miR-1 (63)
High responders and low responders resistance training				
Young males	Muscle biopsy (vastus lateralis)	Skeletal muscle adaptation		miR-29a (28)

*PBMCs = Peripheral blood mononuclear cells; RT-PCR = reverse transcription polimerase chain reaction; HDL = high-density lipoprotein; NYHA = New York Heart Association; IC = intermittent claudication.
†Source: Author's own production.

Table 4
Synthesis of the common findings for each type/modality of exercise—acute and chronic effects (upregulation and downregulation).*†

Studied population	Sample type	Function	Upregulation miRNA (references)	Downregulation miRNA (references)
Endurance				
Intensive exercise or premarathon exercise training				
ER and NER	Plasma (RT-PCR)	Atrial remodeling in athletes	miR-1 (22) and miR-133a (22)	miR-29b (62)
Bicycle ergometer				
Healthy males	Muscle biopsy (vastus lateralis)	Skeletal muscle adaptation	miR-1 (82); miR-133a (82); miR-29b (82); and miR-133b (82)	
Healthy young individuals	Plasma (RT-PCR)	Skeletal muscle adaptation	miR-146a (66)	miR-145 (66)
Bicycle ergometer or home aerobic training and the usual care group received standard treatment without specific instructions on physical activity				
Patients with CKD and healthy control group	Plasma (RT-PCR)	Exercise-induced cardiovascular adaptations	miR-146a (93)	

*ER = elite runners; NER = nonelite runners; RT-PCR = reverse transcription polimerase chain reaction; CKD = chronic kidney disease.
†Source: Author's own production.

pre-exercise levels after 2 weeks of normal daily activity (no exercise training) (65). MiR-1, and -133a significantly increased after a half marathon (39). The role of mir-1 is related to myogenesis, where it promotes the differentiation of myoblasts (16,17). In a study by Mooren et al. (62), the specific correlation of miR-1 and -133a to performance parameters indicated their potential role as biomarkers of aerobic capacity. In addition, this downregulation was associated with improved endurance capacity, $\dot{V}O_2$max, and insulin sensitivity (56). Mueller et al. (63) reported different adaptative effects when elderly subjects were submitted to 2 strength-training protocols: conventional resistance exercise training and eccentric ergometer training after 12 weeks. Both strength protocols led to a reduction in miR-1 expression, accompanied by elevated IGF-1 expression and skeletal muscle mass gain.

Individuals with DM2 associated with morbid obesity exhibited reduced miR-29a after aerobic training of 16 weeks

Table 5
Synthesis of the common findings for each type/modality of exercise—acute, chronic, and acute and chronic effects (no changes in regulation).*†

Studied population	Sample type	Function	No changes in regulation miRNA (references)
Acute effect—resistance: bilateral leg extension and leg press			
Healthy, inactive, young	Serum	Declines in skeletal muscle mass, increased fat mass, and "anabolic resistance"	miR-486 (54) and miR-100-5p (54)
Chronic effect—endurance: bicycle ergometer			
HDL healthy and individuals with CHF‡	Serum (RT-PCR)	Atherogenesis and endothelial dysfunction	miR-221 (80) and miR-222 (80)
Chronic effect—endurance: rowing			
Healthy athletes	Plasma (RT-PCR)	Control and repair of immune function, cell cyte process, development, and growth	miR-133a (6)
Chronic effect—endurance: walking on a treadmill			
Patients with IC	§Peripheral blood (RT-PCR)	Anti-inflammatory	miR-146a (68)
Chronic effect—resistance: resistance training			
Healthy elderly	Plasma and muscle tissue (RT-PCR)	Skeletal muscle adaptation	miR-1 (104)
Type 2-diabetics with morbid obesity	Muscle biopsy (vastus lateralis)	Skeletal muscle adaptation	miR-29a (81)
Acute and chronic—endurance: bicycle ergometer			
Healthy young males	Serum (RT-PCR)	Biomarkers for aerobic capacity	miR-1 (4) and miR-133a (4)
Acute and chronic—endurance: bicycle ergometer or home aerobic training and the usual care group received standard treatment without specific instructions on PA			
Patients with CKD and healthy control group	Plasma (RT-PCR)	Exercise-induced cardiovascular adaptations	miR-145 (93)

*RT-PCR, *r*everse transcription polimerase chain reaction; CKD, chronic kidney disease. PA, physical activity.
†Source: Author's own production.
‡HDLNYHA
§Isolation of total nucleated cells and serum.

(81). According to the authors, the downregulated miR-29a also connected to the vascular development transcriptome network. Among the sample of 17 young males, Davidsen et al. (28) found that miR-29a levels were reduced in low responders to resistance exercise training after 12 weeks, similarly indicating a role in hypertrophy. The miR-29a is involved in protein synthesis; however, the mechanisms involved are not fully known yet (56).

Clauss et al. (22) determined the value of circulating microRNAs as potential biomarkers for atrial remodeling in marathon runners (miRathon study). MiR-1 and -133a increased but showed significant changes in elite runners (ERs) only 24 h after the marathon plasma levels returned to baseline. In ERs, miRNA plasma levels showed a significant correlation with LA diameter suggesting that circulating miRNAs could potentially serve as biomarkers of atrial remodeling in athletes.

Russell et al. (83) have shown that miR-1 and -133a levels were increased in healthy males during a 3-hour period after a single cycling session. Furthermore, the above authors found increased miR-1 and miR-29b levels in the vastus lateralis muscle of healthy males after 10 days of aerobic training on an ergometer bicycle at 75% $\dot{V}O_2$max.

Blood-born miRNA expression pattern have been reported for various human diseases with signatures specific for diseases. Van Craenenbroeck et al. (93) evaluated the acute and chronic effects of exercise in patients with CKD compared with healthy controls. As for the acute effect, miR-146a levels were reduced in patients with CKD. Chronic exercise, by contrast, did not induce any significant changes.

Acute and chronic effects of physical exercise on an ergometer bicycle have also been observed in a study on young healthy individuals (67). Specifically, miR-146a levels were reduced, and miR-145 levels were increased after 1 hour of training.

Similar to the review study of Flowers et al. (36), the studies identified differential expression of miRNAs associated with endurance versus resistance training, miRNAs associated with acute versus chronic exercise, and acute and chronic changes in miRNA expression after extended exercise training programs. The miRNA signature is different according to each stimulus, once reports identify specific circulating miRNA profiles immediately after, hours after, and days after exercise. Thus, the time point of sample collection is crucial to use circulating miRNAs as biomarkers to reflect the effect of exercise (40). Although exercise is widely accepted to have numerous health benefits, there is variability in the frequency, intensity, and duration of exercise required for improved health outcomes. In our review, the frequency, intensity, and duration of exercise varied from 2 to 5 days, moderate and high intensity (evaluated by $\dot{V}O_2$max, $\dot{V}O_2$peak, HRmax, repetition maximum [RM], lactate threshold, and Pmax) for 30–60 minutes each session, respectively. Most studies reported a quasiexperimental design used to assess changes in miRNA expression before and after exercise.

The characteristics of the physical exercise (such as intensity, duration, frequency, modality, and trained level) and age of the practicizer play a central role in the regulation of gene expression, which can occur by different mechanisms (e.g., generation of second messengers, modulation of myogenic regulatory, and epigenetic factors) (56). In the study of Zacharewicz et al. (102), there was difference in the expression of miRNAs between healthy young and older males. In the study of Drummond et al. (33), young males exhibited lower levels after exercise.

Most of the published studies were oriented on the influence of the endurance exercise on the c-miRNAs expression. The recent discovery of miRNAs in human biofluids, including serum, plasma, saliva, sweat, cerebrospinal fluid, urine, and milk, has revealed new possibilities for miRNA research (18). Such ci-miRNAs are actively or passively secreted into the bloodstream where they circulate in association with EVs (microvesicles, exosomes, or apoptotic bodies), proteins (e.g., Argonaute), or high-density lipoproteins (84).

Besides that, circulating miRNAs can have a cellular origin, as in the case of the studies that evaluated monocytes (79), neutrophils (76), and whole blood (30,26). Expression of cellular miRNAs may represent active regulation of genes activated in those cells. By contrast, the origin of miRNAs found in plasma and serum is unknown (10). According to some authors, either serum or plasma have been shown to exhibit a higher concentration of cimiRNA, possibly as a result of the coagulation process or remnant cellular components, respectively (38,59,87,96). These miRNAs may reflect changes in regulation of genes associated with any tissue source. For example, during resistance training, skeletal muscle tissue is damaged, and myocyte miRNAs may be sloughed into the circulation (36).

Circulating miRNAs meet most of the requirements for good biomarkers, such as minimally invasive and easily accessible sample collection, and remarkable stability in body fluids. Fluid type–specific miRNAs may have functional roles associated with the surrounding tissues, and the spectrum and concentration of miRNAs in body fluids vary greatly in addition to show different profiles in pathological states (99), reinforcing their role as non-invasive biomarkers (40). The possible correlation of c-miRNAs regulated by exercise with disease-modified c-miRNA patterns should be also taken into account during developing c-miRNAs signatures as physical performance biomarkers (74).

There is vast evidence that ci-miRNA alterations facilitate the progression of diseases. Likewise, the ability of exercise training to prevent and mitigate diseases is well characterized, so it seems likely that ci-miRNAs may mediate such processes. The few studies detailed here (26,68,72,80,93,101) support to this notion, although more mechanistic approaches are obviously needed, as well as time course and longer term, well-controlled training studies (84). A clinical implication is to enable oriented interventions to promote health.

More studies are needed that compare young, healthy, sedentary, and trained subjects, either cross-sectionally or in longitudinal training studies, to more completely understand the effects of exercise training (84). In our review, some studies were conducted with men and women, but the comparison by sex was not performed (5,60,63,71,72,81,91,93). Some studies found a statistically significant difference between healthy subjects and subjects with disease (72,80,93), athletes and nonathletes (5,22), and healthy young, and older males (33,102).

Most studies reported a quasiexperimental design used to assess changes in miRNA expression before and after exercise, that is, they did not present random distribution of the subjects and neither control groups. From the studies that evaluated the effect of the resistance training, 8 used the 1RM method to determine the maximum strength of the participants (11,33,37,54,70,85,102,104). The 1RM test depends on great human resource direction and high demand in terms of time; however, it is a method often used to measure muscle strength in research and has proven to be a safe and reliable measurement instrument for several populations before to start resistance exercise programs. Most endurance studies reported quantitation of fitness using an objective method (e.g., $\dot{V}O_2$max), which decreases possible bias. All the studies used reverse transcription polymerase chain reaction, the gold standard method for miRNA quantification.

In conclusion, under this perspective, it is safe to assume that miRNA changes, both in muscle tissues and bodily fluids, are presumably associated with the benefits induced by acute and chronic physical exercise. Thus, a better understanding of changes in miRNAs as a response to physical exercise might contribute to the development of miRNAs as therapeutic targets for the improvement of exercise capacity in individuals with any given disease. However, it is important to emphasize that differences in miRNA expression can have a very different meaning when serum/plasma, total blood, total blood cells, or total nucleated cells are examined. Therefore, new studies should be performed to investigate the effect of physical exercise on circulating miRNAs in individuals with pathological conditions, such as neurological diseases, because physical exercise plays an important role in mitigating or delaying the emergence of symptoms and thus grants a certain degree of independence and improved quality of life, emphasizing its consideration as an auxiliary means in traditional therapies. Furthermore, studies should be performed to determine the type of physical exercise, the ideal frequency and duration of sessions, and the adequate intensity and period of the intervention and follow-up to ensure and maintain the benefits related to the effects of exercise on circulating miRNAs.

Practical Applications

Presented results show that miRNA changes, both in muscle tissues and bodily fluids, may distinguish between the variations in the modality, duration, and type of exercise. Circulating microRNA signatures or tissue biopsy may assist, through the available evidence, that athletes and coaches, patients and health professionals in a more precise and accurate way. Particularly, the circulating miRNAs as blood-based biomarkers of exercise response are highly promising because they meet most of the requirements for good biomarkers, such as minimally invasive and easily accessible sample collection, and remarkable stability in body fluids.

The identification of circulating miRNAs signatures in response to a particular exercise modality where they can be used to monitor the response to specific workload regimens, as markers of performance, or as diagnostic tools able to early detect pathophysiological conditions. Also, the evidence found in this study of the information on miRNAs signatures will be of great contribute to the achievement of practical applications in the prevention or therapy of various diseases associated with specific training programs, such the intensity of undesirable consequences of a training program, beneficial effects of exercise training for a particular patient, the full recovery of the athlete health condition, to healthy young and older individuals, and the individualized exercise prescription.

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